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===== TEACHER SCRIPT: MHF4U UNIT 6 -

PROBLEM SOLVING SESSION Student: Rigden Total Slides: 27

HOW TO USE THIS SCRIPT: Keep this text file open on a second monitor. The presentation file itself contains zero teacher notes to ensure it is 100% student-safe. Guide Rigden through the interactive tabs (Think, Hint, Solution) on each slide using these prompts.

SLIDE 1: Title Slide [SAY] Welcome back, Rigden! We've covered the theory of Rates of Change. Today is a pure problem-solving session. [SAY] We have 25 distinct challenges ahead of us, ranging from basic intervals to complex velocity applications. [ASK] Do you have your paper, pencil, and calculator ready? The calculator is critical today!

LEVEL 1: FUNDAMENTALS (Slides 2 - 6)

SLIDE 2: Q1 - AROC Fundamentals [ASK] How do we find the Average Rate of Change algebraically? [EXPECT] The slope formula: $[f(b) - f(a)] / (b - a)$. [SAY] Exactly. Calculate $f(4)$ and $f(1)$ carefully.

SLIDE 3: Q2 - Secant Slope from Table [ASK] Does a secant line represent an average rate or an instantaneous rate? [EXPECT] Average rate of change (AROC). [SAY] Right. Just find the standard slope between $t=0$ and $t=4$ using the table.

SLIDE 4: Q3 - Rational AROC [SAY] A rational function works the exact same way. [ASK] Calculate $g(4)$ and $g(1)$, then apply the slope formula.

SLIDE 5: Q4 - AROC Direction [ASK] If the slope of a secant line is exactly zero, what does that line look like? [EXPECT] Perfectly horizontal. [SAY] Correct. This means the y-values at the start and end of the interval must be identical.

SLIDE 6: Q5 - Following Interval Setup [SAY] We are switching to IROC estimation now. [ASK] What is the difference quotient formula? [EXPECT] $[f(a + h) - f(a)] / h$. [SAY] Set it up for this specific problem using $h = 0.01$. Do not solve it yet.

LEVEL 2: INTERMEDIATE (Slides 7 - 12)

SLIDE 7: Q6 - IROC of a Quadratic [SAY] Now we execute the calculation. [ASK] Use $h = 0.001$. Walk me through calculating $f(3.001)$. [TIP] Remind him not to round anything until the very final answer! Keep all decimal places.

SLIDE 8: Q7 - IROC of a Cubic [SAY] Same process, but a cubic function. [ASK] Be very careful with the negative sign in front of the 2. Calculate the difference quotient.

SLIDE 9: Q8 - Radical IROC [SAY] A square root function. [ASK] Setup the difference quotient for $f(2.001)$ and $f(2)$.

SLIDE 10: Q9 - Rational IROC [SAY] A rational function. [ASK] Calculate $4 / (4.001 - 2)$. Tell me the full decimal before you subtract.

SLIDE 11: Q10 - Exponential IROC [SAY] Exponential functions grow very fast. [ASK] Apply the exact same $h=0.001$ logic to the exponent t .

SLIDE 12: Q11 - Turning Points [ASK] Without doing any math, picture a curve hitting a local maximum (a peak). If you draw a tangent line right at the top, what does it look like? [EXPECT] It is

horizontal. [SAY] Exactly. And what is the slope of a horizontal line? (Zero!).

LEVEL 3: ADVANCED (Slides 13 - 18)

SLIDE 13: Q12 - Velocity Word Problem [SAY] Word problems in Unit 6 are almost always about velocity. [ASK] Instantaneous velocity means finding the IROC. Set it up for $t=2$ using $h=0.001$. [EXPECT] Expect a negative answer because it is a downward parabola ($-4.9t^2$).

SLIDE 14: Q13 - Population Growth [ASK] The year is 2015, and the model starts in 2010. What 't' value are we using? [EXPECT] $t = 5$. [SAY] Correct. Find the IROC at $t=5$.

SLIDE 15: Q14 - Preceding Interval [SAY] We usually use following intervals ($+h$). This asks for a preceding interval. [ASK] How does the formula change if we are stepping backwards? [EXPECT] We use $f(x)$ minus $f(x-h)$.

SLIDE 16: Q15 - Centered Interval [SAY] A centered interval uses points on both sides of our target. [ASK] If we go 0.01 forward and 0.01 backward, what is the total distance (the denominator) between our two points? [EXPECT] 0.02.

SLIDE 17: Q16 - Unit Conversion Trap [SAY] Draining water. [ASK] Calculate the IROC. If the volume is decreasing, what sign should your answer have? [EXPECT] Negative.

SLIDE 18: Q17 - Tangent Equation (Part 1) [ASK] This is a conceptual check. If you calculate an IROC of 6, what does that number represent physically on the graph? [EXPECT] The slope (m) of the tangent line.

LEVEL 4: EXAM LEVEL (Slides 19 - 26)

SLIDE 19: Q18 - Tangent Equation (Part 2) [SAY] You know the slope is 6. To find the equation $y = mx + b$, you need a point (x, y) . [ASK] How do you find the exact y -coordinate when $x=3$? [EXPECT] Plug 3 into the original function $f(x) = x^2$.

SLIDE 20: Q19 - Rates on Trig Functions [SAY] You must ensure your calculator is in RADIANS mode for this! [ASK] Calculate $\sin(\pi/4 + 0.001)$ minus $\sin(\pi/4)$, all over 0.001.

SLIDE 21: Q20 - Comparing AROC and IROC [SAY] We want to see how the average rate compares to the exact rate at the midpoint. [ASK] Calculate the AROC on $[1, 3]$, then the IROC at $x=2$. Which is larger?

SLIDE 22: Q21 - Unknown Constant [SAY] We are working backwards. We know the AROC is 15. We don't know 'k'. [ASK] Set up the AROC formula, leave 'k' in there, set it equal to 15, and solve algebraically.

SLIDE 23: Q22 - Interpreting Rates Graphically [SAY] Look at a downward opening parabola (an upside-down U). [ASK] Where on that curve is the slope the most positive (steepest going up)? [EXPECT] At the very beginning (left side) before it starts to flatten out at the peak.

SLIDE 24: Q23 - IROC of a Linear Function [SAY] A trick question! $f(x) = 5x - 2$. [ASK] Do we need the difference quotient to find the IROC of a straight line? [EXPECT] No, the slope of a line is constant. It is always 5.

SLIDE 25: Q24 - IROC at an Asymptote [ASK] Why can you not calculate the IROC at exactly $x=2$ for $f(x) = 1/(x-2)$? [EXPECT] The function is undefined there. There is a vertical asymptote. No point = no tangent line.

SLIDE 26: Q25 - True or False [ASK] If a function is strictly decreasing, will its secant and tangent lines always have negative slopes? [EXPECT] True. Decreasing means going down-to-the-right, which is a negative slope.

SLIDE 27: Wrap Up [SAY] Incredible endurance today, Rigden. You successfully navigated the Unit 6 problem set. [SAY] Review these strategies, specifically the difference quotient setup, before your Unit 6 exam.